

Interactive Image Processing: Design Of An AI Assisted Human Computer Interaction Logic Using Gesture Recognition Model Based On Logical Open CV Scheme

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Abstract—One method that can recognize hand gestures in real-time video is hand gesture recognition. A specific field of study is used to categorize the hand gesture. One of the difficult tasks in this study involving two main problems is the design of the hand gesture recognition system. The first step is to identify the hand. Making a sign that works for just one hand at a time is another challenge. The focus of this system is on developing a computer vision system that can detect, identify, and understand hand gestures, even when faced with tough circumstances such as variations in scale, orientation, position, and attitude. A variety of gestures, including numerical and sign language ones, must be implemented into this system for the development of this study endeavour to be successful. There are a lot of potential uses for this field, which provides a more organic and easy method for humans to interact with robots and system interfaces without the usage of additional hardware. Therefore, developing systems that can recognize individual human gestures and utilize them to transmit data or operate devices is the major objective of gesture recognition research as it pertains to Human-Computer Interaction (HCI). This necessitates real-time gesture identification and lightning-fast, very robust hand detection for vision-based hand gesture interfaces. Using an AI algorithm, this study introduces a solution for real-time gesture detection called the Logical Gesture detection Model (LGRM). It is general enough to be applied in many different kinds of human-computer interactions. To verify that the suggested model is effective, it is cross-validated with the traditional method known as the Cascading Gesture Classification Model (CGCM).

Keywords—Image Processing, Artificial Intelligence, AI, Human Computer Interaction, Gesture Recognition, OpenCV, LGRM, Gesture Classification, CGCM

I. INTRODUCTION

Everyday life includes systems that can recognize hand gestures in real-time videos, and one of them is hand gesture recognition [1]. A specific field of study is used to categorize the hand gesture. The project's goals include, among other things, the design of a system for hand gesture recognition. Computer vision has several essential and fundamental problems, one of which is the recognition of hand motions. Modern developments in media and information technology have made possible human-computer interaction (HCI) systems that perform hand processing tasks, including gesture recognition and hand detection [2][3].

Hand processing systems begin by identifying and localizing the hand in the webcam's live feed [4]. Because of variations in size, position, orientation, and attitude, hand identification is problematic. The variation is further amplified by the fact that the room's lighting is of varying intensities. There are often many stages involved in hand detection and gesture identification, including picture capture, pre-processing, feature extraction, and gesture recognition. Acquiring images requires taking still pictures from a movie using a camera and saving them frame by frame. Color filtration, smoothing, and thresholding are all steps in the picture pre-processing pipeline that are applied to the collected photos. Gesture recognition uses feature extraction to identify hand gestures, while feature extraction

uses hand images to extract characteristics such as hand outlines [5][6]. The development of systems that can recognize individual gestures and use them as inputs or outputs is a key focus of gesture recognition research [7]. While hand posture refers to the hand's static structure, gestures denote the hand's dynamic movement, and both the spatial and temporal domains must be accounted for when modelling gestures [8].

One example is sign language; it's the most natural means of communication for the deaf community, but they still struggle to communicate with hearing people. Similar to how a spoken language's vocabulary consists of words, a sign language's vocabulary comprises of signs. Different countries use different grammars in sign languages; hence there is no universally accepted norm. For several reasons, including but not limited to making it possible for young children to engage with computers (via sign language recognition), the development of fast and accurate methods for translating sign language into text or speech has been a primary focus of Sign Language Recognition (SLR) systems. The two main schools of thought when it comes to hand gesture identification are vision-based and data glove techniques [9]. In order to develop a system that can recognize posture and gestures for usage in real-time applications, this study focuses on building a vision-based method. Since vision-based hand gesture recognition systems provide a more natural and easy method for humans to communicate with computers, they were the primary emphasis of the research.

When interacting with a computer, hand gestures are a natural and reliable way to transmit data. While conventional input devices like keyboards, mice, joysticks, and touch screens do the job, the current system will feature an interface that is more suited to desktop or laptop computers, allowing users to perform hand gestures with the use of data gloves or a webcam to capture an image of their hand. Capturing and analyzing hands is the first stage in this gesture recognition process. In contrast, the vision-based approach only requires a camera, thereby detecting the real human-computer interaction without the need for any additional hardware. Constant backdrop, sometimes human, and lighting are additional obstacles for this method. Here, in addition to the recognition approaches, we detail the many procedures and algorithms that make up this system [10].

II. RELATED STUDY

Digital mental health solutions that are efficient, user-friendly, cost-effective, as well as adaptive have been designed and developed in part via human-computer interface (HCI) [11]. However, there are worries about quality and safety since HCI has not been properly integrated into technology advancements. There is great promise for digital platforms and AI to enhance mental health care as well as suicide prevention services' capacity

for prediction, detection, coordination, and treatment. Websites and mobile applications powered by AI are mostly used for self-help along with guided cognitive behavioral therapy (CBT) for mental health issues including anxiety and depression. Outdated, overburdened, or nonexistent mental healthcare systems may benefit from interactive AI's real-time screening and treatment capabilities. Challenges in using AI in mental healthcare include issues with accessibility, effectiveness, dependability, usability, security, ethics, appropriate training and education, and socio-cultural adaption. Digital phenotyping, immersive technologies, apps, and real-time machine learning algorithms are promising areas to watch out for. Improved and more rapid human-machine interaction and automation, more stringent evaluations of efficacy, and the adjunctive use of blended, hybrid, or phased care are all necessities. Assisting in the digital therapeutic alliance, identifying and resolving disparities in mental health treatment and suicide prevention, and designing and developing practical applications are all areas that might benefit from human-computer interaction (HCI) modelling [11].

Product design in this age of fast progress must address not only functional and technical requirements, but also the psychological and emotional demands of consumers [12]. Traditional human-computer interaction interfaces were ill-equipped to handle modern users' demands; so, psychological design material must be included into modern HCIs. Consequently, this study proposes a research methodology for human-computer interaction interface design grounded on design psychology with the aim of enhancing the user experience. This study examines a few real-world issues with the design of human-computer interaction interfaces via the lens of design psychology, looking at these issues through the lenses of human behavior and perception. Afterwards, we look into the elements that Chinese users care about while using a human-computer interface and the percentage of distinct sensations in the human-computer information that people get. New needs for human-computer interaction interface design have emerged, according to the research. In order to usher in a new age of design psychology-based research on human-computer interaction interfaces, it is necessary to compile the relevant experimental data and findings [12].

The purpose of this research is to ascertain how human-computer interface design influenced the evolution of mobile phone technology [13]. Descriptive analysis was the method used by the researcher. In addition to highlighting the benefits of cell phones, such as improved social interaction and faster application loading times, the results also highlight the drawbacks, like disturbed sleep and increased weariness, of this technology. The study also identifies HCI as a field that studies computer technology, with a focus on user interfaces, and the common problems that HCI faces, such as the fact that HCI designers have a hard time figuring out how to make sure their designs have good HCI. The study's authors urged more research into

mobile phones and their respective platforms, usability testing of software and websites, and the proper qualification and training of users with cognitive impairments so that they can make better use of these tools [13].

It is widely believed that information and communication technology may help developing nations' socioeconomic situations improve [14]. The growth of underserved populations has been made possible, according to studies, in several ways by the improved telecommunications infrastructure. As an example of an actual ICT application, digital library systems address many unfulfilled demands in research institutions, scientific communities, and new innovations while also providing better solutions overall. The parallel database system has evolved into the digital library system's primary instrument for effective data processing as technology has progressed. Thus, the coordinator, the collection machine, and the query processor can finish the distribution, load, and maintenance operations by coordinating their communication in the parallel environment. This has a high degree of efficiency, saves a lot of time, helps the digital library meet user needs effectively, and satisfies the data performance requirements of the digital library. Additionally, the key problem in the parallel algorithm has been solved. The efficiency and effectiveness of this parallel approach are shown by the experimental findings [14].

Important input techniques in Human-Computer Interaction (HCI), including as gesture and voice recognition, have seen extensive application in VR in recent years [15]. Specifically, voice recognition and gesture recognition have accomplished groundbreaking research advances thanks to the fast growth of deep learning, AI, and other computer technologies. The primary search engines used for this project are Google Scholar and the Web of Science literature database. Nearly a thousand researches were chosen based on HCI and deep learning-related keywords including "intelligent HCI," "speech recognition," "gesture recognition," and "natural language processing." Afterwards, after screening over 500 research technique studies throughout the course of five years (2019–2022), 100 papers were ultimately chosen to constitute the research material of this work. The present state of the HCI intelligent system is examined first, followed by a summary of the achievements in HCI regarding gesture and voice interaction, and finally, the benefits of deep learning are chosen for further study. The next section lays out the fundamentals of gesture interaction before reviewing the history of interaction between gesture and voice recognition. Also included are descriptions of some typical uses for voice recognition and gesture recognition in conjunction with one another. We conclude by looking at the state of the art in human-computer interaction as it relates to natural language processing. The results demonstrate the extensive use of intelligent HCI and deep learning in areas such as intelligent robot guidance, emotion identification, voice

recognition, and gesture recognition. In related scientific disciplines, many other recognition algorithms were suggested and experimentally confirmed. Impressive recognition accuracy was attained when contrasted with interactive approaches that did not include deep learning. Improving user interfaces is a key function of context in voice-supported Human-Machine Interfaces (HMIs). Action recognition may be made much more accurate and precise by combining long short-term memory networks with convolutional neural networks. That is why additional sectors will be part of HCI's future application area, and even better things are on the horizon [15].

III. METHODOLOGY

Everything revolves around the shape of the hand. Any line or surface that connects all the continuous points along its boundaries and has no straight or flat parts is said to have contours. The thresholding step is carried out prior to this one. Using a webcam, we capture a basic picture of the hand in RGB format. Producing a binary picture that details the hand's outline requires some preparatory work. In this case, the contour and convex-hull of the contour are determined using the binary picture. If you have a collection of points and want to draw a closed figure around them, the shortest possible outline is the convex-hull of that set. As if every point on the hand's contour falls inside the convex hull's bounds, it's drawn on both sides. The effect is the same as wrapping one's hand in a folded paper container. A flaw in the hand gesture becomes apparent when convex hulls are drawn on both sides of the hand's contour, fitting sets of contour points. To build the hull and include all of the contour points within, it makes advantage of maximum locations. When a convexity defect is present, it produces a vector of values. The following figure Fig.1 shows the flow diagram of the LGRM method.

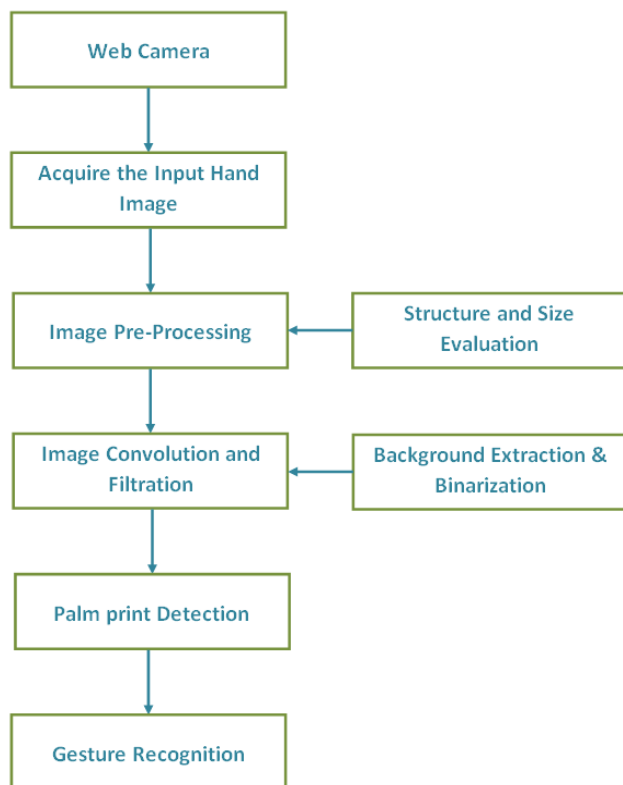


Fig.1 Flow Diagram of LGRM Method

This method uses data collected from the convex hull to identify the palm's location and the hand's shape to identify the fingertip's location. This system's computing cost is minimal since the computation geometry algorithm is conducted. The proposed method Logical Gesture Recognition Model (LGRM), an alternative to the current Cascading Gesture Classification Model (CGCM), increases the reliability of hand identification. The following figure Fig.2 shows the proposed flow diagram.

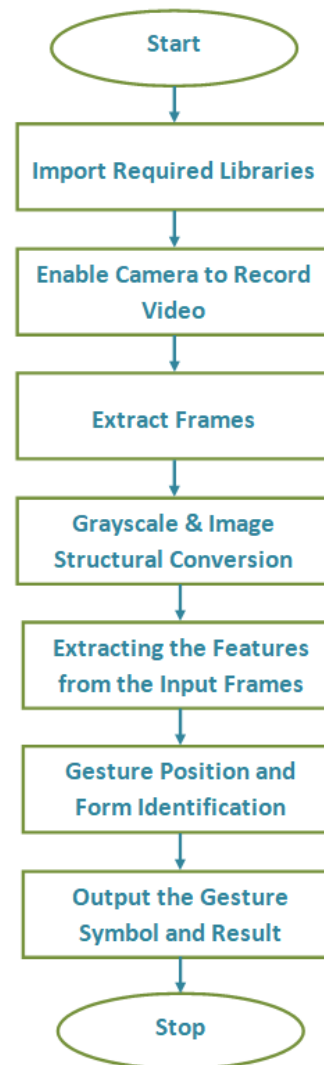


Fig.2 Proposed Flow Diagram

Data gathering and pre-processing, data representation or feature extraction, and classification or decision-making are the three main components of any gesture recognition system design. The user's hand is detected and tracked by the system, which then extracts relevant hand information by segmenting the hand from the video picture. Using these attributes, we can determine the user's gesture. Prior to using the acquired instance vector for classification, the characteristics pertaining to a static gesture must be normalized. However, in order to classify a dynamic gesture, the resulting hand path is first tagged using a preset alphabet, producing a discrete vector of labels. This vector is then translated back to its origin and utilized for classification. An algorithm takes in each detected gesture and uses it to construct the command sequence; that is, it adds up all the gestures it receives until it finds a preset sequence described in the Command Language.

(i) Capturing Images: In the first stage, we acquired the RGB picture (frame by frame) with the use of a camera and solely hand motions.

(ii) Pre-Processing: We have extracted the relevant region (also called the "Region of Interest") from the video stream rather than the whole frame in order to reduce the amount of time needed for calculations. Thus, we transform the area of interest into a grayscale picture by using image processing to transform the original color photos into grayscale. Once the processing is finished, the images are returned to their original color space.

(iii) Hand Area Segmentation: This step enables the development of the system's functioning by removing superfluous data from the video stream, which is crucial for any process involving hand gestures identification. The first method relies on skin color; it's simple but effective, and it takes into account environmental light and the background's characteristics to identify the hand in a picture. In the second method, we use a hand shape to our advantage by capitalizing on the convexity concept while detecting hands. When learning to recognize a hand signal, the position of the hand is crucial.

(iv) Extracting the Contour: The visible border of the item (the hand) in the picture serves to outline the contour. Important in form analysis and object recognition methods, the contour may also be a wave linking spots with comparable color values.

(v) Identifying and Extracting Features: A collection of peaks that encompasses the hand area called the convex hull. Here, we need to establish the convex set principle, which states that any line connecting any two points on the hull is completely included inside it. Once the gesture has been identified, the desired action may be carried out. Movement recognition is an evolving process. Return to the first stage to accept another picture for processing, etc., after carrying out the specified instruction from the gesture.

The three main parts of the framework are the Camera, Detection, and Interface components. The following figure Fig.3 shows the system architecture.

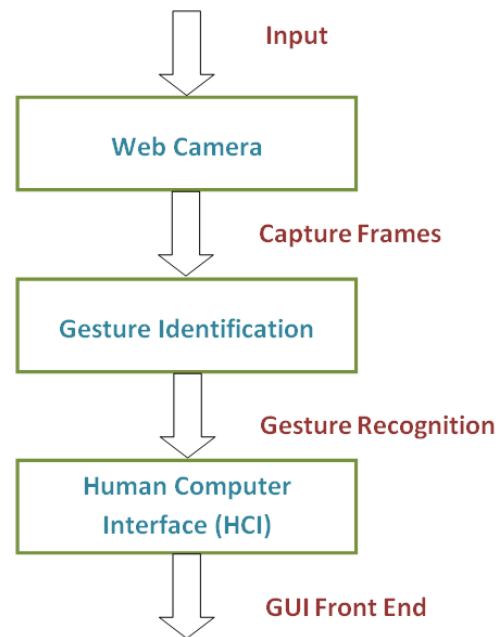


Fig.3 System Architecture

(a) The Camera Module: This is responsible for the actual taking of pictures, which it transmits to the detection module in the form of frames after interacting with various image markers. Cameras, data gloves, and hand belts are the most common ways to capture and identify input.

(b) The Detection Module: This is responsible for processing images. Color conversion, noise reduction, thresholding, and other image treatment procedures are applied to the camera module's output before contour extraction.

(c) The Interface Module: Tailoring the identified hand gestures to the corresponding actions is the responsibility of this module. After that, the appropriate application receives these instructions.

IV. RESULTS AND DISCUSSIONS

There are typically three main steps in most gesture recognition algorithms. Identifying specific things inside digital still photos or moving movies is the primary goal of the object detection phase. At this point, improving the identification accuracy requires solving several environmental and picture difficulties in order to extract the contours or areas with precision. Issues with noise, contrast, resolution, and fluctuating brightness are common in images. These issues can be efficiently addressed by implementing better environmental conditions and camera technologies. Nevertheless, it is challenging to regulate the operation of the gesture recognition system in a real-world setting or when it is being developed into a product. So, to build an adaptable and strong gesture recognition system, the image processing approach is the way to go for solving these picture challenges. Realization of objects is the second phase. In order to recognize the motions, the detected items

on the hands are recognized. Currently, the majority of study is focused on the following areas: effectively selecting classifiers; and distinct characteristics. The third step is to decipher the user's commands or actions by analyzing their successive movements. For improved gesture identification and to address issues with previous methods like the Cascading Gesture Classification Model (CGCM), a new algorithm dubbed the Logical Gesture Recognition Model (LGRM) is suggested.

Figure 4 shows the input picture for the suggested method, and Figure 5 shows the pre-processing result for the same scheme.

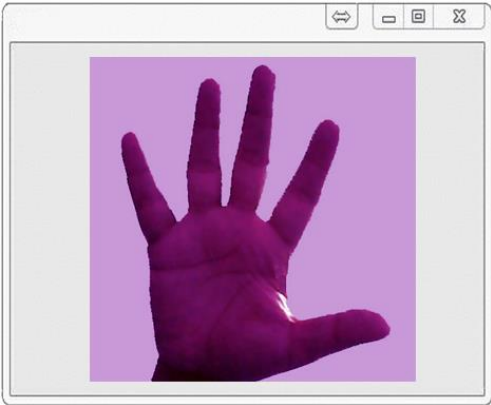


Fig.4 Input Image



Fig.5 Pre-Processing

The results of the feature extraction stage of the suggested method are shown in Figure 6, and the results of the edge identification stage of the suggested scheme are shown in Figure 7.

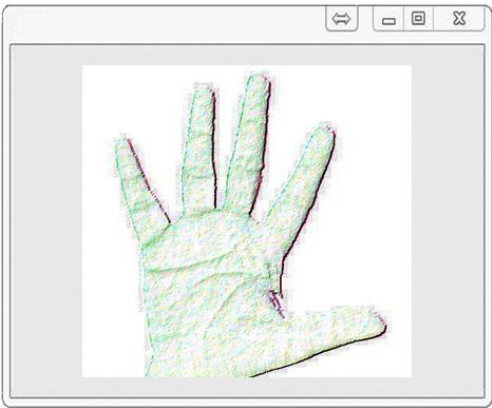


Fig.6 Feature Extraction

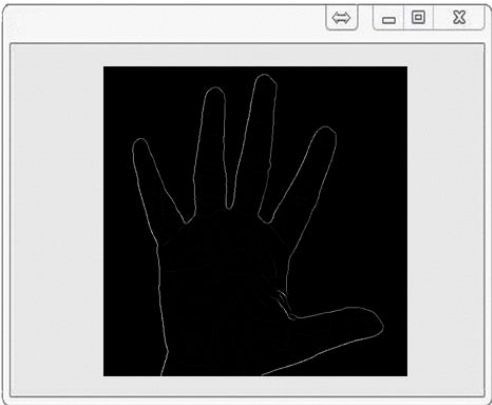


Fig.7 Edge Recognition

In order to clearly evaluate the efficiency of the proposed scheme, it is cross-validated with the conventional Cascading Gesture Classification Model (CGCM) scheme. Following this, Figure 8 shows the classification stage outcome of the proposed approach, and Figure 9 shows the analysis of the prediction accuracy of the proposed scheme. Table-1 provides a descriptive representation of the same.

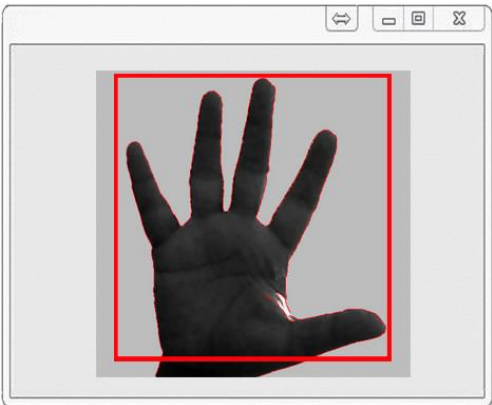


Fig.8 Classification

TABLE-1: PREDICTION ACCURACY ANALYSIS

Epochs	CGCM (%)	LGRM (%)
100	86.34	98.64
125	86.52	96.52
150	84.31	97.47
175	85.36	96.39
200	87.61	95.49
225	86.52	97.67
250	85.45	96.72
275	86.11	98.31
300	86.13	97.07
325	86.15	97.05
350	86.17	97.03
375	86.20	97.01
400	86.22	96.99

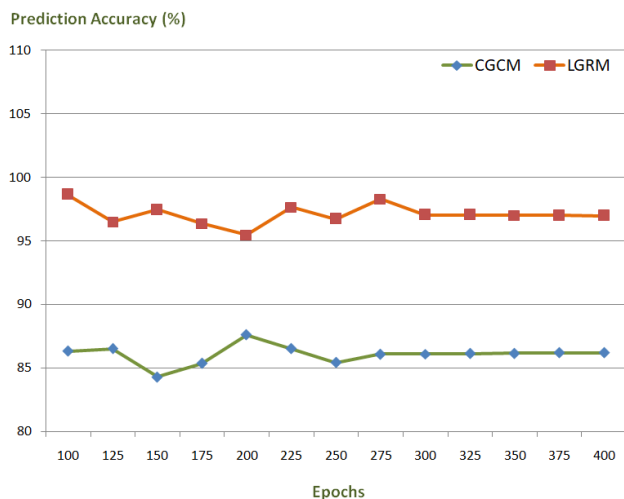


Fig.9 Prediction Accuracy Analysis

V. CONCLUSION

According to the findings of the system, the development of hand gesture recognition using Python and OpenCV is made possible by using the theories of hand fragmentation and the hand recognition system that employs the Haar-cascade classifier. This is evident from the results of the system. In conclusion, this system has accomplished a number of objectives: first, it has developed a comprehensive system for detecting, recognizing, and interpreting hand gestures through computer vision using Python and OpenCV; second, it has generated sufficient examples of hand gestures and sign languages to satisfy the requirements of its name; and third, it has successfully implemented these objectives. In the future, this system will be able to carry out a greater number of motions, which will make it simpler for users with a variety of skin tones and

palm sizes to carry out a greater number of functions. In the current technique, the right hand is only used inside a certain region of interest (ROI) in order to produce motions. In light of this, the desired technique enhancement can entail the utilization of both hands in order to carry out a variety of computer-based indicators. You have the option of using procedures that involve background subtraction for even better outcomes. It is anticipated that a Graphical User Interface (GUI) would be implemented in the near future, which will enable users to effortlessly turn motions into signs and numbers, and vice versa.

REFERENCES

- [1] Jiaming Wang, et al., "Research Trends of Human-Computer Interaction Studies in Construction Hazard Recognition: A Bibliometric Review", *Sensors*, <https://doi.org/10.3390/s21186172>, 2021.
- [2] M. S. et al, "Smart Face Detection and Recognition in Illumination Invariant Images using AlexNet CNN Compare Accuracy with SVM," (ICIEEM), London, United Kingdom, 2022, pp. 572-575, doi: 10.1109/ICIEEM54221.2022.9853078.
- [3] H Hristov, et al., "Generations of human-computer interactions: evolution, tendencies and perspectives", *Journal of Physics: Conference Series*, doi:10.1088/1742-6596/2339/1/012009, 2022.
- [4] G. R et al, "An Effective Copyright Management Principle using Intelligent Wavelet Transformation based Water marking Scheme," (ACCAI), Chennai, India, 2022, pp. 1-7, doi: 10.1109/ACCAI53970.2022.9752516.
- [5] Daniela Quiñones, et al., "Understanding the customer experience in human-computer interaction: a systematic literature review", *PeerJ Comput Sci*, doi:10.7717/peerj-cs.1219, 2023.
- [6] Christos Troussas, et al., "A User-centric System for Improving Human-Computer Interaction through Fuzzy Logic-based Assistive Messages", 17th International Conference on Web Information Systems and Technologies, 2021.
- [7] C. Mansoor, G. Vishnupriya, A. Anand, S. Vijayakumar, G. Kumaran and V. Samuthira Pandi, "A Novel Framework on QoS in IoT Applications for Improving Adaptability and Distributiveness," 2023 *International Conference on Computer Communication and Informatics (ICCCI)*, Coimbatore, India, 2023, pp. 1-6, doi: 10.1109/ICCCI56745.2023.10128257.
- [8] Mahmoud Hawamdeh, et al., "Human-Computer Interaction (HCI) Interventions for People with Disabilities", <https://doi.org/10.32388/N2S82Q>, 2023.
- [9] Hari Singh, et al., "Object Acquisition and Selection in Human Computer Interaction Systems: A Review", *International Journal of Intelligent Systems and Applications in Engineering*, <https://doi.org/10.18201/ijisae.2019151248>, 2019.
- [10] Sharefa Murad, et al., "CHR vs. Human-Computer Interaction Design for Emerging Technologies: Two Case Studies", *Advances in Human-Computer Interaction*, <https://doi.org/10.1155/2023/8710638>, 2023.
- [11] Samuthira Pandi, V., Singh, M., Grover, A., Malhotra, J., Singh, S, "Performance analysis of 400 Gbit/s hybrid space division multiplexing-polarization division multiplexing-coherent detection-orthogonal frequency division multiplexing-based free-space optics transmission system", *International Journal of Communication Systems*, 2022, 35(16), e5310..
- [12] Yina Liu, "Human-Computer Interface Design Based on Design Psychology", *International Conference on Intelligent Computing and Human-Computer Interaction*, DOI: 10.1109/ICHCI51889.2020.00009, 2020.
- [13] Nidal Al Said, et al., "Assessment of Acceptance and User Experience of Human-Computer Interaction with a Computer Interface", *International Journal of Interactive Mobile Technologies*, DOI:10.3991/ijim.v14i11.13943, 2020.
- [14] Jun Cao, "Design and Implementation of Human-Computer Interaction System in Parallel Digital Library System Based on Neural Network", *Scientific Programming*, <https://doi.org/10.1155/2021/9921551>, 2021.
- [15] Zhihan Lv, et al., "Deep Learning for Intelligent Human-Computer Interaction", *Appl. Sci.*, <https://doi.org/10.3390/app122211457>, 2022.